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CarICL: Mitigating causal hallucinations to enhance event causality identification

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ABSTRACT

Event Causality Identification (ECI) is a critical and challenging Natural Language Processing (NLP) task. Despite Large Language Models (LLMs) offer the potential for ground-breaking achievements through In-Context Learning (ICL), they still demonstrate significant weaknesses in ECI, as evidenced by series causal hallucinations. We argue that this is due to three major shortcomings of ICL for ECI: (1) conventional retrieval methods fail to ensure sufficient causal similarity between input queries and ICL demonstrations; (2) the lack of explanations of causal relationships in demonstrations leading to poor ICL effectiveness; and (3) limited non-causal knowledge in LLMs causing a misalignment between ICL and human causal cognition. In this paper, we propose *CarICL* to address the aforementioned issues by: (1) incorporating causality-aware representations in demonstration retrieval; (2) enriching the demonstrations with cause-and-effect reasoning; and (3) aligning LLMs with human causal cognition through causality preference optimization. Experimental results show that *CarICL* outperforms state-of-the-art baselines on three widely used sentence-level ECI benchmarks.

1. Introduction

Events are fundamental linguistic components that convey both high-level semantic meaning and commonsense knowledge in natural language. Causality denotes a specific relationship between events where the occurrence of one event directly or indirectly leads to the occurrence or alteration of another. In recent years, Event Causality Identification (ECI) has emerged as a crucial task, focusing on determining whether a causal relationship exists between given event pairs within a sentence. ECI has been widely used in various Natural Language Processing (NLP) tasks (Le et al., 2025; Zhang et al., 2025; Zixuan et al., 2025). Effectively identifying event causality is challenging because it is often constrained by the problem of causal data scarcity (Cheng et al., 2024).

Early works (Zuo et al., 2021a,b, 2020) have demonstrated the effectiveness of using data augmentation techniques with Pre-trained Language Models (PLMs), such as BERT (Devlin et al., 2019) and RoBERTa (Liu et al., 2019), for the ECI task. Although these works have made notable progress, their advancement heavily relies on external knowledge bases, which limits their usability in real-world applications (Cheng et al., 2024). The emergence of Large Language Models (LLMs), particularly GPT-series decoder-only models such as GPT-3 (Brown et al., 2020), has introduced a new paradigm known as In-Context Learning (ICL). With the help of ICL, LLMs outperform PLM-based methods even when provided with only a few demonstrations, significantly alleviating data scarcity issues in numerous NLP tasks (Chan et al., 2024; Pu et al., 2023b; Qin et al., 2023; Wang et al., 2023). Despite the overall promising

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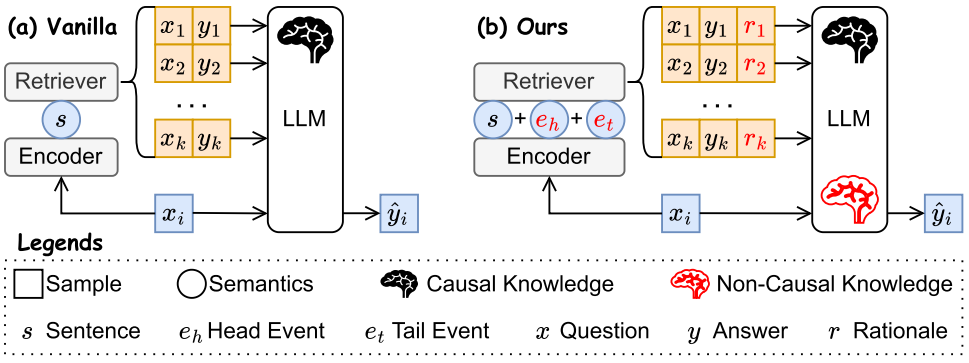


Fig. 1. Overview of how In-Context Learning (ICL) is applied to the Event Causality Identification (ECI) task.

performance of LLMs, the utilization of ICL for the ECI task is still suboptimal. As shown in Fig. 1(a), the LLM is prompted with a few demonstrations retrieved from the training data and the query input. The LLM then generates the corresponding prediction based on this combined input. Recently, several studies (Chen et al., 2024b; Gao et al., 2023; Huang et al., 2024; Li et al., 2024) have applied vanilla ICL to the ECI task and found that LLMs exhibit serious causal hallucination issues, often assuming non-existent causal relationships between events (as illustrated in Fig. 2(a)). As a result, the performance of LLMs with ICL still significantly underperforms PLM-based methods. In this paper, we identify three primary causes of causal hallucinations when applying ICL to the ECI task.

Firstly, in vanilla ICL, the demonstrations retrieved using conventional retrieval methods exhibit low causal similarity. As shown in Fig. 2(b), demonstrations are selected through a k-Nearest Neighbors (kNN) search based on sentence embeddings. Unfortunately, such methods solely rely on sentence embeddings, which focus more on the overall semantic similarity of sentences and pay less attention to the causal relationships between event mentions. Therefore, the subsequent ICL inference lacks high-quality demonstrations for performing causal reasoning.

Secondly, the lack of explanations for question-answer mappings in demonstrations reduces the effectiveness of vanilla ICL. Without explicit reasoning steps, LLMs may learn only surface-level associations rather than the underlying causal logic — unfortunately, this problem is further exacerbated because causal relations can be expressed in many different ways. As a result, they rely solely on implicit reasoning, ignoring the fact that recent studies have shown explicit reasoning can greatly improve model performance and reasoning ability (Chen et al., 2025).

Finally, the limited non-causal knowledge in LLMs leads to a misalignment with human causal cognition. The primary reason for this may be that natural language contains numerous causal relationship descriptions, indicated by causal cue words such as “lead to” and “therefore”. However, it generally does not specify which events are not causally related. Since the causal reasoning abilities of LLMs primarily stem from pre-training on massive amounts of natural language corpora, the reporting bias between causal and non-causal events enables them to acquire substantial causal knowledge, but only limited non-causal knowledge (Kiciman et al., 2024).

Research objectives. Given the above challenges, this study aims to explore three key research questions: (1) How to retrieve high-quality demonstrations for ICL that are causally relevant rather than merely semantically similar? (2) How to guide LLMs to perform explicit cause-and-effect reasoning instead of relying on shallow pattern matching? and (3) How to align LLMs with human causal cognition by integrating both causal and non-causal knowledge?

We are motivated to address the above limitations with the goal of mitigating both the causal data scarcity and causal hallucination problems. To this end, we present **CarICL**: a novel **C**ausality-**a**ware demonstration **r**etrieval **I**n-**C**ontext **L**earning framework, where we fully exploit the potential of 7B-parameter LLMs through task-aware demonstration retrieval, step-by-step reasoning, and preference

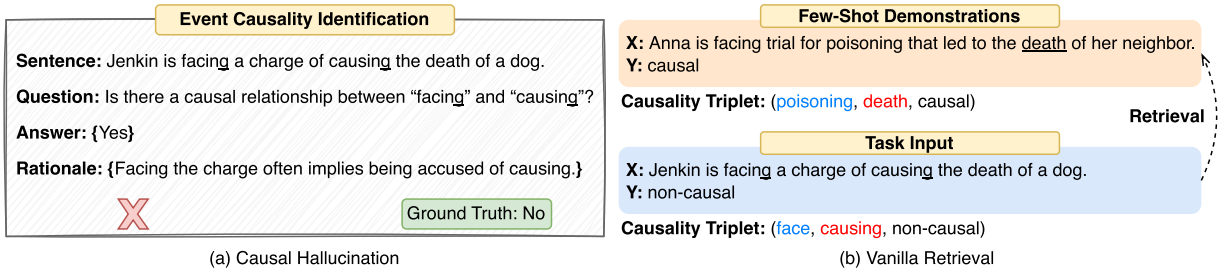


Fig. 2. (a) An illustration of causal hallucination. Content requiring LLM responses is enclosed within the symbols “{” and “}”. (b) Retrieval without considering the causality-aware triplet results in noisy demonstrations.

optimization. To address the first issue, we propose a causality-aware demonstration retrieval method. This method contains more causality-specific information than sentence-level semantics alone, thereby significantly enhancing the causal similarity of retrieved ICL demonstrations. To address the second issue, we propose a cause-and-effect reasoning method that injects explicit reasoning logic into demonstrations. This method enables ICL not only to learn causal patterns from surface-level features but also to perform more advanced step-by-step explicit reasoning. To address the third issue, we propose a causality preference optimization method that introduces non-causal knowledge into LLMs without destroying existing causal knowledge. This method significantly enhances the alignment of LLMs with human causal cognition and achieves the elimination of the reporting bias by injecting comprehensive causal knowledge.

Overall, we make the following contributions:

- We propose a novel ICL framework (CarICL) that identifies event causalities while mitigating causal data scarcity and causal hallucination problems. To the best of our knowledge, this is the first study that fully leverages LLM and ICL techniques to significantly outperform PLM-based methods on three sentence-level ECI benchmarks.
- We propose (1) a causality-aware retrieval method to search causally similar demonstrations, (2) a cause-and-effect reasoning method to enhance the effectiveness of ICL, and (3) a causality preference optimization method to align LLMs with human causal cognition.
- Experiments and analyses demonstrate that CarICL (1) not only outperforms 175B LLMs with just 7B parameters but also surpasses state-of-the-art baselines; (2) effectively addresses low-resource scenarios and generalizes to other tasks by leveraging LLM and ICL techniques; and (3) mitigates causal hallucinations by retrieving high-quality demonstrations.

2. Related work

2.1. Event causality identification

Event-centered causal reasoning plays a pivotal role in numerous downstream NLP applications (Cheng et al., 2024), enabling deeper semantic understanding and more accurate inference capabilities. Existing research has explored various approaches to enhance machine causal reasoning capabilities. Cao et al. (2021) integrated external structural knowledge to simulate human causal reasoning mechanisms. Mu and Li (2023) introduce counterfactual reasoning to address the ambiguity of causal signals by explicitly measuring context-keyword relationships and eliminating event-pair biases. Su et al. (2025) propose a knowledge-enhanced method that leverages external knowledge and concept-level event relations to strengthen event-knowledge associations and uncover causal relationships. Although these PLM-based methods perform well with sufficient annotated data and external knowledge, they perform poorly in low-resource scenarios.

To the best of our knowledge, our paper is the first study to mitigate the need for extensive training data and external knowledge by employing LLMs and ICL, offering a simple yet effective approach that demonstrates particularly strong performance in low-resource settings.

2.2. In-context learning

2.2.1. Demonstration selection

Recent works have established adaptive selections of few-shot demonstrations, which substantially enhance the ICL performance compared to static demonstration sets (Liu et al., 2022; Qin et al., 2024). Most approaches leverage sentence embeddings for retrieval, particularly by utilizing sentence encoders from PLMs. For instance, Liu et al. (2022) introduced KATE, a k-nearest neighbors framework for unsupervised retrieval of in-context demonstrations. Su et al. (2023) introduced a demonstration selection framework that integrates graph-based representations with confidence scoring to optimize the diversity of in-context examples. Wu et al. (2023) developed an information-theoretic approach to showing the efficacy of using LLM output scores as unsupervised metrics for demonstration selection. Despite the NLP community has introduced numerous innovative methods for ICL demonstration selection, limited attention has been devoted to optimizing demonstration selection specifically for the ECI tasks.

To the best of our knowledge, we take the first step in addressing this gap by introducing prompt-based event mention information for in-context demonstration selection in the ECI task.

2.2.2. Instruction formatting

A common method for formatting demonstrations involves concatenating examples $(x_i, y_i)_{i=1}^{k}$, where x_i are questions and y_i are the corresponding answers. However, this approach is insufficient for tasks requiring complex reasoning, where the mapping from questions to answers is not easily inferred from a limited set of demonstrations (Wei et al., 2022b; Zhang et al., 2023c; Zhou et al., 2023). Therefore, Wei et al. (2022b) propose the CoT prompting technique to enhance problem-solving and comprehension by introducing intermediate reasoning steps between questions and answers. This technique involves manually writing rationales r and incorporating them into corresponding demonstrations, forming new demonstrations $\{(x_i, r_i, y_i)\}_{i=1}^{k}$. Zhang et al. (2023c) introduce Auto-CoT, which autonomously generates reasoning process r to elicit effective CoT responses, addressing the limitation of manual CoT prompt engineering. Zhou et al. (2023) propose least-to-most CoT prompting, which decomposes complex problems into simpler sub-problems, solving them sequentially with prior solutions assisting in subsequent ones. Despite previous CoT approaches have demonstrated substantial advances in commonsense and numerical reasoning through automated rationale generation, they face

unique challenges in the ECI task. Specifically, the ECI task involves event pairs that may exhibit multiple potential relations, making reasoning generation unfocused without explicit guidance from gold labels.

In this paper, we propose a novel cause-and-effect reasoning method that appends the reasoning process after answers to provide causal explanations. This approach substantially differs from previous works that position the reasoning process between questions and answers as an intermediate step. Formally, our proposed formatting can be represented as $\{(x_i, y_i, r_i)\}_{i=1}^{k}$.

2.2.3. Model warmup

Recent studies show that ICL capabilities can be significantly enhanced through specialized training prior to ICL inference. A simple yet effective approach involves introducing a continual training stage between pre-training and ICL inference, commonly referred to as "model warmup" (Dong et al., 2024). For example, Wei et al. (2023) introduced symbol tuning, replacing natural language labels (e.g., "positive/negative sentiment") with arbitrary symbols (e.g., "foo/bar"), to learn question-answer mappings from context. Meanwhile, several studies have highlighted the importance of instruction tuning in enhancing LLMs' downstream task performance. For instance, (Wei et al., 2022a) propose FLAN to improve LLMs' instruction-following ability, significantly boosting both zero-shot and few-shot ICL performance. Wang et al. (2022b) and Chung et al. (2024) further scaled up instruction tuning by incorporating over one thousand task-specific instructions. Despite previous studies have shown that instruction tuning can improve few-shot ICL performance, almost none of them discuss how task-aware demonstration retrieval-based instruction fine-tuning impacts the performance in the subsequent inference stage.

In this paper, we propose a novel ICL model warmup method that not only leverages task-aware demonstration retrieval to enhance instruction-following abilities but also introduces causality preference optimization to align LLMs with human causal cognition.

2.3. Causal hallucination

Causal reasoning plays a pivotal role in many NLP tasks. Recent systems focused on causal reasoning have primarily relied on fine-tuned Pre-trained Language Models (PLMs) like BERT (Devlin et al., 2019) and RoBERTa (Liu et al., 2019). However, these models' causal reasoning capabilities are dependent on supervised training with large volumes of annotated data. Recently, large language models (LLMs), such as ChatGPT (Hurst et al., 2024), have demonstrated impressive performance across a variety of NLP tasks (Chan et al., 2024; Qin et al., 2023; Wang et al., 2023). Despite these advancements, recent studies (Chen et al., 2024b; Gao et al., 2023; Huang et al., 2024; Li et al., 2024) have identified a major challenge: LLMs are prone to causal hallucination, resulting in significantly lower performance and reliability on causal reasoning tasks compared to PLM-based methods. Specifically, causal hallucination is defined as the tendency of LLMs to assume causal relationships between events, regardless of whether such relationships actually exist. This phenomenon is typically assessed by observing high recall but low precision in LLMs' outputs — indicating that many non-causal event pairs are incorrectly labeled as causal.

In this paper, we aim to address these challenges by improving not only overall performance (as measured by F1-score) but also reliability (as reflected in the gap between recall and precision).

3. Methodology

In this section, we present a detailed technical description of our proposed approach. Our goal is to mitigate both the causal data scarcity and causal hallucination problems in the ECI task.

3.1. Task definition

The ECI task is structured as a binary classification problem. The ECI datasets are formally represented as $\mathcal{Z} = (\mathcal{X}, \mathcal{Y})$, which are usually human annotated from sentences. The symbol $\mathcal{X}$ denotes the collection of sentences, and symbol $\mathcal{Y}$ represents the corresponding set of causality labels. Each individual sentence $x \in \mathcal{X}$ contains at least two event mentions, i.e., a head event mention and a tail event mention. We use the symbols $e_h \in x$ and $e_t \in x$ to represent the head and tail event mentions, respectively. The objective of the ECI task is to predict the relationship $y \in \{0, 1\}$ between a pair of event mentions within a given sentence x. Specifically, $y = 0$ indicates that no causal relationship exists, while $y = 1$ signifies that a causal relationship does exist.

3.2. Overview

Fig. 3 presents an overview of our proposed CarICL framework, which consists of two stages: Stage 1, Causality Preference Optimization (Section 3.4); and Stage 2, ICL inference. **Firstly**, we conduct preference optimization on the causality preference dataset. This dataset is constructed using our proposed prompt construction method (Section 3.3). This method consists of causality-aware demonstration retrieval (Section 3.3.1) and cause-and-effect reasoning (Section 3.3.2). Notably, in this process, the queries are iterated from the training set, and the demonstrations are also retrieved from the training set. This process can be viewed as a form of ICL warmup. **Secondly**, we evaluate the causality preference-optimized LLM on the test set, which corresponds to ICL inference. The prompt construction method is the same as in Stage 1. The difference from Stage 1 is that, in this process, the queries are iterated from the test set, while the demonstrations are still retrieved from the training set. Next, we will provide a detailed explanation of each component.

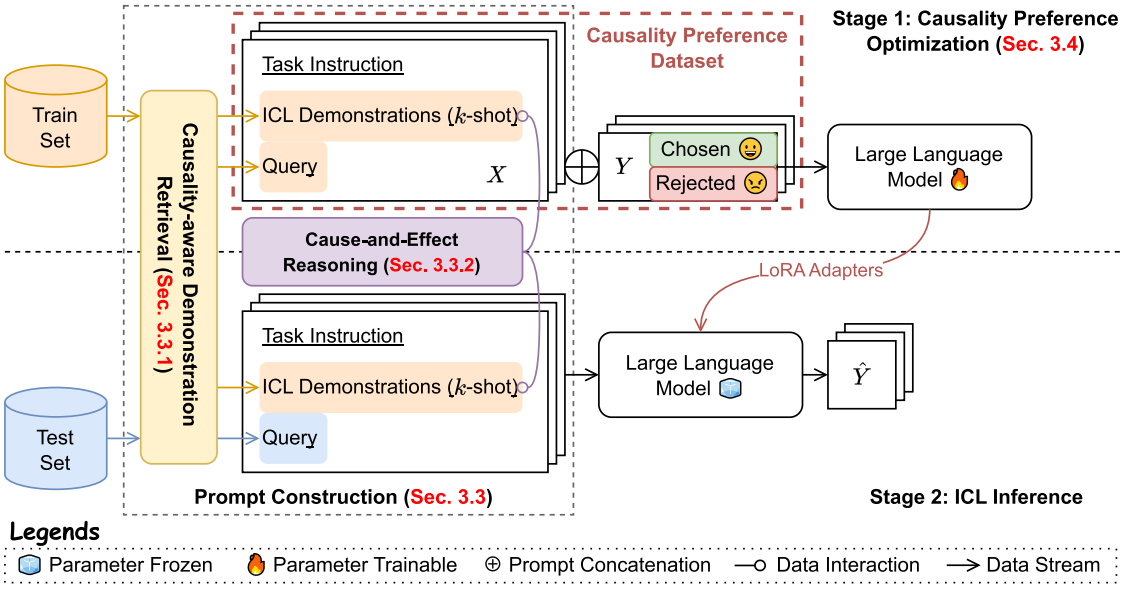


Fig. 3. An overview of our proposed CarICL framework, which consists of two stages. First, we conduct Causality Preference Optimization (Section 3.4) on the training set, which can be viewed as a form of ICL warmup (Section 2.2.3). Second, we evaluate the optimized LLM on the test set, which corresponds to ICL inference.

3.3. Prompt construction

By leveraging natural language instructions, LLMs have demonstrated remarkable abilities as general-purpose task solvers (Brown et al., 2020; Wei et al., 2022b; Zhang et al., 2023c; Zhou et al., 2023). The effectiveness of their performance heavily depends on the quality of the prompt guiding the model. In the ECI task, recent LLM-based approaches utilize open-ended templates for label verbalization, resulting in low accuracies (Chen et al., 2024b; Gao et al., 2023; Huang et al., 2024; Li et al., 2024). Specifically, based on the given query, if the LLM generates "yes", it indicates a causal relationship. If the model generates "no", it signifies that no causal relationship exists in the corresponding query. Inspired by recent studies (Ouyang et al., 2022; Wang et al., 2022b; Zhang et al., 2023b), we think that open-ended templates fail to elicit strong ECI capabilities in LLMs due to the low prevalence of open-ended templates in instruction-tuning datasets.

To address this limitation, we propose a new prompt construction method that aligns the ECI task with the multiple-choice question-answering task, as this type of task appears much more frequently in most instruction-tuning datasets. Specifically, by casting the input sentence as a question and the possible options as multiple-choice answers, LLMs can perform the ECI task by selecting the option representing the correct causal relation. Fig. 4 illustrates this method.

Task instructions $\mathcal{I}_1$. We provide a description of the ECI task and the expected behavior of the model. To better elicit the instruction-following abilities of LLMs, which have dramatically improved in recent models, we reformulate the ECI task as a multiple-choice question-answering format prompt. By integrating the given head and tail event mentions (e_h and e_t) into the prompt templates and casting the possible causal relations as multiple-choice options, we enhance the model's ability to understand the ECI task. Additionally, our method allows LLMs to generate only an answer index instead of the verbalized causal relation, as popular in previous works.

ICL demonstrations $\mathcal{D}$. Given a query input x_{query}, we first utilize causality-aware retrieval (described in Section 3.3.1) to obtain a k-shot demonstration set that is causally similar to the query. Next, we enrich each demonstration (x_i, y_i) with r_i, derived from cause-and-effect reasoning (described in Section 3.3.2), to construct a new set of (x_i, y_i, r_i) as follows:

$$\mathcal{D} = \{(x_1,\ y_1,\ r_1), \ldots, (x_k,\ y_k,\ r_k)\}. \tag{1}$$

Query input x_{query}. Similar to the instruction and demonstrations, we also provide the query input x_{query} to the LLM. Notably, during the causality preference optimization stage (described in Section 3.4), x_{query} is iterated from the training set. In contrast, during the ICL inference stage, x_{query} is iterated from the testing set. Then, the LLM is expected to select the correct answer y_{query} from the instruction $\mathcal{I}$.

In summary, the likelihood of the correct answer y_{query} is determined by a scoring function f applied to the entire input sequence:

$$\mathbf{P}(y_{\text{query}} \mid x_{\text{query}}) \triangleq f_{\text{LLM}}(\mathcal{I}_1,\ \mathcal{D},\ x_{\text{query}}). \tag{2}$$

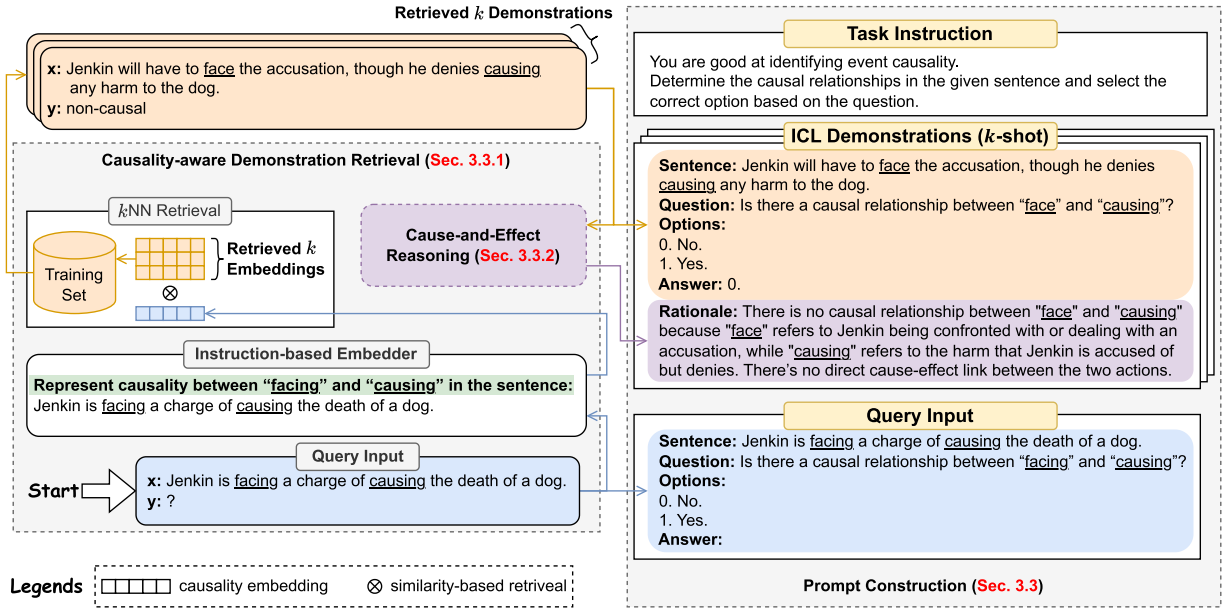


Fig. 4. An illustration of our proposed prompt construction method. Given an query input (x, y), we first use the proposed causality-aware retrieval method (as described in Section 3.3.1) to search for high-quality demonstrations from the training set. Next, we incorporate cause-and-effect reasoning (as described in Section 3.3.2) into each demonstration. Finally, the above contents are combined into a prompt to make a prediction.

The final predicted label $\hat{y}$ is selected as the candidate answer with the highest predicted probability:

$$\hat{y} = \arg\max_{y_{\text{query}} \in \mathcal{Y}} \mathbf{P}(y_{\text{query}} \mid x_{\text{query}}). \tag{3}$$

3.3.1. *Causality-aware demonstration retrieval*

In the ICL inference stage, demonstrations that are closer to the query input in the embedding space often lead to more consistent and robust performance (Liu et al., 2022). Recent works (Chen et al., 2024b; Gao et al., 2023; Huang et al., 2024; Li et al., 2024) show that the kNN algorithm is a practical approach for retrieving high-quality examples from the training set as few-shot demonstrations for each query input. Since the kNN algorithm relies on the embedding space to represent both the query input and training examples, researchers suggest using PLMs or other enhanced models (Feng et al., 2025) to generate more effective sentence embeddings. However, solely using sentence embeddings for kNN retrieval presents a significant drawback in the ECI task. Specifically, the ECI task pays more attention to pairwise event mentions (i.e., e_h and e_t) and the causality between them, which often diverges significantly from the overall semantic meaning of the entire sentence. This divergence results in low causal similarity in retrieved demonstrations.

In this paper, we propose a novel retrieval method to provide more effective representations for better retrieval quality. We argue that conventional sentence embedding methods often struggle to adapt to different retrieval intents for two reasons. First, solely relying on the original context is insufficient for retrieving causally similar demonstrations. Second, vanilla sentence encoders often fail to deeply understand the query intent. Considering the importance of event mentions in the ECI task and the emphasis on causal similarity in retrieval, we propose not only reconstructing the context by incorporating event pair information and instructions but also replacing vanilla sentence encoders with an instruction-following embedder. For example, our goal is to represent the causal relationship between two event mentions ("facing" and "causing") in the given context:

Original Context

Jenkin is facing a charge of causing the death of a dog.

We reconstruct the original context with an instruction (highlighted in bold) that provides explicit descriptions of our intent and emphasizes the event mentions along with the context:

Instruction-based Context

Represent the causality between "facing" and "causing" in the sentence:
Jenkin is facing a charge of causing the death of a dog.

This approach captures the sentence's semantic meaning and instructs the embedding model to focus on event pair-centered causal relationships. Formally, the causality embedding h for a given sentence x and events e_h and e_t, based on the instruction $\mathcal{I}_2(\cdot,\cdot)$, is

computed as follows:

$$h = \mathbf{E}(\mathcal{I}_2(e_h\ ,\ e_t)\ \oplus\ x), \tag{4}$$

where $\mathbf{E}(\cdot)$ denotes the instruction-following embedder, and the symbol $\oplus$ represents the prompt concatenation operation. With this approach, causality embeddings can be pre-computed offline, enhancing training and inference efficiency at the expense of increased storage requirements.

3.3.2. Cause-and-effect reasoning

Many CoT studies have demonstrated significant advancements in commonsense and numerical reasoning tasks by enabling automatic step-by-step reasoning to solve complex questions (Chen et al., 2025; Qin et al., 2023; Wei et al., 2022b). Recent researchers have further shown that LLMs demonstrate a degree of proficiency in reasoning about sentence-level temporal, causal, and discourse relations (Chan et al., 2024). However, in the ECI task, two event mentions can potentially hold multiple relationships (Wang et al., 2022a). For example, "facing" and "causing" can have coreference, temporal, causal, discourse, or other types of event relationships. Therefore, the reasoning generated by LLMs may be unfocused if it lacks interaction with the gold label.

To address this limitation, we propose generating reasoning logic for each demonstration using ChatGPT, based on the corresponding sentence, event mentions, and gold label. In this paper, we first construct a query prompt based on the given demonstration. The prompt template is highlighted in bold, and the expected reasoning generated by ChatGPT is enclosed in {…} notation:

Query Prompt

You are good at explaining causal relationships.
Explain the causal relationship step-by-step as briefly as possible based on the given sentence, question, and answer.
Sentence: Jenkin is facing a charge of causing the death of a dog.
Question: Is there a causal relationship between "facing" **and** "causing"**?**
Answer: No.
Rationale: {…}

Next, we prompt ChatGPT to generate cause-and-effect reasoning based on the labeled causal relationship between the event mentions in the sentence. The resulting response is shown below:

Response from ChatGPT

"Facing" means being accused, while "causing" means being responsible.
Being accused does not necessarily mean actual responsibility.
Therefore, there is no causal relationship between "facing" and "causing".

Finally, we enrich the demonstrations by incorporating the cause-and-effect reasoning generated by ChatGPT. Formally, the cause-and-effect reasoning r for a given sentence x and causal relationship y, based on the query prompt $\mathcal{I}_3(\cdot,\cdot)$, is generated as follows:

$$r = \mathbf{LLM}(\mathcal{I}_3(e_h\ ,\ e_t)\ \oplus\ x\ \oplus\ y), \tag{5}$$

where $\mathbf{LLM}(\cdot)$ refers to a powerful LLM, such as ChatGPT in this paper.

3.4. Causality preference optimization

With the increase in the number of model parameters and the volume of training data, LLMs have begun to demonstrate the ability of ICL, which means learning from just a few demonstrations within the context (Brown et al., 2020). Many studies have demonstrated that LLMs are capable of performing a wide range of complicated tasks through ICL, including solving complex mathematical reasoning problems (Chan et al., 2024; Qin et al., 2023; Wei et al., 2022b). Despite ICL holds great potential, it exhibits notable limitations when applied to the ECI task, as demonstrated by its frequent inaccuracies (Chen et al., 2024b; Gao et al., 2023; Huang et al., 2024; Li et al., 2024). We argue that natural language contains numerous explicit expressions of event causal relationships in various forms (e.g., "because", "result in"), but it rarely specifies which events are not causally related or explains why they lack a causal connection. As a result, this reporting bias can cause LLMs to have incomplete causal knowledge. Furthermore, the limited non-causal knowledge may undermine the alignment between LLMs and human causal reasoning.

To address this limitation, we propose a novel Causality Preference Optimization (CPO) method to integrate non-causal knowledge into LLMs without modifying their existing causal knowledge. It consists of two stages: generating preference data (as described in Section 3.3) and fine-tuning the model (as described in this section). Leveraging our generated preference data, we can either conduct supervised fine-tuning with the annotated labels or implement preference optimization using the preference pairs.

Supervised fine-tuning. For Supervised Fine-Tuning (SFT), we optimize the LLM π_θ by minimizing the average negative log-likelihood loss, formulated as:

$$\mathcal{L}_{SFT} = -\log P_\theta(y \mid x) \tag{6}$$

$$= -\frac{1}{|y|}\log \pi_\theta(y \mid x) \tag{7}$$

Table 1
Dataset Statistics. The words OOD, ID, CI, CT, and CD denote Out-of-Distribution, In-Distribution, Class-Imbalance, Cross-Task, and Cross-Domain, respectively. The symbol † denotes entities rather than events.

Dataset	Topic	Event mention	Event pair	Causal pair	Causal rate (%)	Evaluation
ESL	22	5334	7805	1770	22.7	OOD
ESL♣						ID
CTB	–	6948	7608	318	4.2	CI
SET8	–	9994†	10717†	1331†	12.4	CI, CT
HP	–	171	827	190	23.0	CI, CD

$$= -\frac{1}{|y|}\sum_{i=1}^{|y|}\log \pi_\theta(y_i \mid x,\ y_{<i}), \tag{8}$$

where y denotes the LLM's output given a prompt x, and we use the exponentiated average log-likelihood $P_\theta(y|x)$ to represent the likelihood of output y given prompt x in the form of probability.

Preference optimization. Alternatively, we can perform preference optimization to strengthen the tendency toward correct causality predictions while minimizing the likelihood of incorrect ones. From various available preference optimization methods, we adopt the monolithic Odds Ratio Preference Optimization (ORPO) algorithm (Hong et al., 2024) based on its demonstrated empirical effectiveness. Specifically, the ORPO algorithm incorporates a novel odds ratio loss function that directly optimizes the model by minimizing the negative log-odds ratio between preferred (y-win, denoted as y_w) and dispreferred (y-loss, denoted as y_l) outputs:

$$\mathcal{L}_{OR} = -\log \sigma\left(\log \frac{\text{odd}_\theta(y_w \mid x)}{\text{odd}_\theta(y_l \mid x)}\right). \tag{9}$$

Here, the symbol σ denotes the sigmoid function, and the term $\text{odd}_\theta(y|x)$ serves as a metric to quantify the relative likelihood of the model generating output y versus not generating it:

$$\text{odd}_\theta(y \mid x) = \frac{P_\theta(y \mid x)}{1 - P_\theta(y \mid x)} \tag{10}$$

$$= \frac{\pi_\theta(y \mid x)}{1 - \pi_\theta(y \mid x)}. \tag{11}$$

The ORPO final objective integrates SFT and OR losses, with a hyper-parameter β regulating their relative importance:

$$\mathcal{L}_{\text{ORPO}} = \mathcal{L}_{\text{SFT}} + \beta \cdot \mathcal{L}_{\text{OR}}. \tag{12}$$

In our implementation, we use the correct causal relationship corresponding to query x as y_w, and the incorrect causal relationship corresponding to query x as y_l.

4. Experiment

4.1. Datasets

Following previous works (Gao et al., 2024; Li et al., 2024), we evaluate the proposed CarICL on sentence-level ECI datasets. Table 1 presents the statistical results for these benchmarks.

EventStoryLine v0.9 (ESL) (Caselli & Vossen, 2017) is a widely used event causality dataset for the ECI task(Gao et al., 2024; Li et al., 2025; Liu et al., 2023). The dataset is primarily gathered from various news websites and features 22 topics, such as natural disasters, killings, accidents, and trials. Consistent with prior work (Gao et al., 2019), we exclude aspectual, causative, perception, and reporting event mentions, as these are rarely annotated with causal relationships. After preprocessing, the dataset spans 22 topics with 5334 event mentions and 7805 intra-sentence event pairs, including 1770 (22.7 %) causal relationships. Following the standard data partitioning approach of previous studies (Gao et al., 2019; Liu et al., 2020), the last two topics in ESL are designated as the development set, while the remaining 20 topics are used for 5-fold cross-validation. To ensure consistency in the distribution of causal relationships between the training and testing sets during cross-validation, we apply stratified sampling based on the label characteristics of causal relationship quantities. Under this data partitioning strategy, sentences are sorted by topic IDs, resulting in cross-topic splits between the training and test sets. The distribution gap between the training and test sets provides a valuable opportunity to evaluate the cross domain generalization capabilities of various models.

ESL♣ represents an alternative partitioning of the ESL dataset, commonly adopted in prior sentence-level ECI studies (Hu et al., 2023; Li et al., 2024; Man et al., 2022; Shen et al., 2022). Unlike the ESL dataset, which organizes data based on topic IDs, the ESL♣ dataset randomly shuffles the data based on sentence IDs. This random shuffling ensures that the distributions of the training, testing, and development sets are more consistent, as all three sets include samples covering all topics. To preserve consistency in the number of causal relationships across the training and testing sets, stratified sampling is employed during cross-validation based on the distribution of causal labels. This partitioning scheme enables experiments that better showcase a model's ability to

identify causal relationships within topic-centric sentences, aligning more closely with real-world application scenarios. In practical applications, many causal event pairs primarily appear within topic-specific data distributions, as the types of event relationships frequently correlate with topics (Hu et al., 2023). Inspired by this observation, we explore two distinct dataset partitioning strategies: cross-topic partitioning (ESL) and random-topic partitioning (ESL♣), to evaluate model effectiveness more comprehensively and thoroughly.

Causal-TimeBank (CTB) (Mirza & Tonelli, 2014) is a widely used sentence-level dataset for the ECI task. This dataset emphasizes explicit causality, where causal links are clearly indicated by causal signals or causative verbs. It allows models for detailed analyses of how causal relations interact with temporal relations. It includes 7608 intra-sentence event pairs and 6948 event mentions, of which only 318 (4.2 %) are annotated with causal relations, highlighting the sparsity of causality and the class imbalance problem in this dataset. Following the standard data partitioning approach used in previous studies (Liu et al., 2020; Zuo et al., 2021a), we perform 10-fold cross-validation on the CTB dataset. To maintain consistency in the number of causal relationships between the training and testing sets, stratified sampling is employed during cross-validation based on the distribution of causal labels.

SemEval-2010-Task-8 (SET8) (Hendrickx et al., 2010), collected through pattern-based web searches, serves as a valuable benchmark for classifying semantic relations between entity nominals. It focuses on identifying semantic relations between entity pairs within sentences. Over time, it has become a widely used sentence-level dataset for evaluating cross-task adaptation capabilities in related ECI studies (Ding et al., 2024; Liu et al., 2020, 2023). Specifically, it involves training on **event** causality corpora and testing on **entity** causality corpora. Each example is a sentence in which a pair of entities and their relation are annotated. The SET8 dataset comprises 10,717 annotated examples and is divided into 8000 training examples and 2717 test examples, making it one of the largest resources for **entity-centric** relation analysis. It includes nine semantic relations, selected to cover a broad range of common and practically relevant relations, such as Cause-and-Effect, Instrument-and-Agency, and Entity-and-Origin. Among the nine types of entity relations annotated in the SET8 dataset, we focus exclusively on causal relations, treating all other types as non-causal. Notably, this dataset is imbalanced, with non-causal relations occurring more frequently than Cause-and-Effect relations (12.4 %).

Hahn-Powell (HP) (Hahn-Powell et al., 2016) is a dataset for identifying causality among **biochemical** interactions, constructed by extracting events from 500 biomedical articles in PubMed. It is designed to support the identification of causal precedence relations between event pairs, thereby transforming isolated biochemical events into ordered mechanistic steps. In this paper, we utilize this dataset to train and evaluate model performance in the biomedical domain. The corpus comprises 916 high-quality annotated event pairs and was labeled by two domain-expert linguists. For experimental evaluation, stratified 10-fold cross-validation is typically used (Akkasi & Moens, 2021). In most causality studies, the labels are collapsed into three classes (E1-precedes-E2, E2-precedes-E1, and Others) to focus exclusively on causality identification. Notably, the corpus is imbalanced: fewer than 23 % of the annotated pairs express causal precedence, while the majority are non-causal, posing a challenge for supervised models.

4.2. Evaluation metrics

Following previous studies (Gao et al., 2019; Liu et al., 2020; Zuo et al., 2021a), we employ Precision (P), Recall (R), and the F1-score (F1) as our evaluation metrics to assess the performance of models in identifying causally related event pairs. The formulas for these metrics are as below:

$$P = \frac{|MP \cap GT|}{|MP|}, \qquad R = \frac{|MP \cap GT|}{|GT|}, \qquad F1 = \frac{2 \cdot P \cdot R}{P + R + \epsilon}.$$

Here, MP denotes the set of causally related event pairs predicted by the model, while GT represents the set of causally related event pairs annotated in the ground truth dataset. To ensure numerical stability, a small positive value, $\epsilon = 1 \times 10^{-9}$, is added to the denominator of the F1 formula. Precision evaluates the model's prediction quality by calculating the ratio of true positives to all predicted positives. Conversely, Recall gauges the model's prediction coverage by determining the ratio of true positives to all actual positives. The F1-score, as the harmonic mean of Precision and Recall, provides a comprehensive measure that balances both quality and coverage, making it especially valuable for imbalanced datasets. In this paper, **the F1-score is used as the main evaluation metric**, as it effectively captures the trade-off between precision and recall, and serves as a comprehensive measure of model performance.

We evaluate model performance not only using above conventional metrics but also introduce new metrics such as Δ and #Params to further demonstrate the effectiveness of our method. Since recent research (Gao et al., 2023; Zou et al., 2025) uses the difference between P and R to evaluate causal hallucination, we also adopt the metric $\Delta = |P - R|$, defined as the absolute difference between P and R, to reflect causal hallucination. Notably, Δ is not intended to replace the F1-score or serve as an isolated performance indicator. Instead, it functions as a supplementary metric to quantify the model's tendency toward causal hallucination, particularly when comparing models with similar F1-scores. Additionally, the #Params metric is used to assess the total number of parameters in the backbone network. We introduce two new metrics (Δ and #Params), where lower values (↓) better align with our motivations.

4.3. Implementation details

Following previous studies (Gao et al., 2024; Li et al., 2024), we use the development set of the ESL dataset to tune the hyperparameters for our proposed CarICL model. we utilize BGE-Large-EN-v1.5[1] as the instruction-following encoder (Section 3.3.1) to

[1] https://huggingface.co/BAAI/bge-large-en-v1.5

encode both the ICL demonstrations and the query inputs. The encoder's parameters are frozen during both pre-training and testing. We first ensure that both the proposed causality preference optimization (Section 3.4) and the ICL inference stages use the same number of demonstrations for training and inference. Then, we set $k = 8$, selected from the range $\{0, 2, 4, 6, 8, 10\}$, which not only achieves strong results on the development set of the ESL dataset but also better accommodates constraints such as the input length limitations of LLMs and computational efficiency.

We use OpenAI's official API[2] to perform the proposed cause-and-effect reasoning (Section 3.3.2). We select the "gpt-3.5-turbo-0125" engine and set the temperature parameter to 0 to minimize randomness. Since the temperature parameter fixed at 0.0 to ensure deterministic outputs, we conduct a single round of reasoning text generation across all datasets.

We use Qwen-2.5-7B-Instruct (Yang et al., 2024b) as the backbone model of CarICL, employing ORPO (Hong et al., 2024) for preference optimization and AdamW (Loshchilov & Hutter, 2019) as the default optimization algorithm during fine-tuning. The fine-tuning is configured with a per-device batch size of 8, and gradients are accumulated over 4 steps. Following the post-training configuration of Qwen-2.5 (Yang et al., 2024b), the model is fine-tuned for a single epoch using a learning rate of 7e−7 and a weight decay of 0.1. The cosine learning rate scheduler is employed with a warmup phase covering 0.05 of the overall training steps. Training is conducted in bfloat16 precision to balance computational efficiency and numerical stability. Additionally, the LLM is loaded in 4-bit precision and enhanced with nested quantization via the bitsandbytes library. Instead of updating all the model parameters, Parameter-Efficient Fine-Tuning (PEFT)[3] is used to adapt only a small subset of parameters. In our implementation, LoRA (Hu et al., 2022) is configured with a rank of 64, a alpha of 128, and a dropout rate of 0.05. The adaptation is applied to all linear layers within the model, ensuring a broad yet efficient update across the overall neural network. For testing, we selected the optimal model checkpoint based on its F1-score on the development set of the ESL dataset.

All experiments were conducted on a Linux server equipped with Intel Xeon Gold 6342 CPUs (2.80 GHz), 256 GB of RAM, and NVIDIA A100 80 GB GPUs. We used Ubuntu 22.04, CUDA 12.2, and PyTorch 2.5 for training and inference.

4.4. Comparison with state-of-the-art models

We compare our approach with state-of-the-art ECI models, which can be roughly categorized into five types: "Data Augmentation", "Context Generation", "Representation Learning", "Structure Knowledge", and "Prompt Tuning". Moreover, we also compare our approach with LLM-based methods to demonstrate that our proposed model has mitigated the causal hallucinations of LLMs.

4.4.1. Data augmentation

KnowDis (Zuo et al., 2020) expands the training data with the assistance of a lexical knowledge base and distant supervision. This approach is motivated by the idea that a sentence containing an event pair with a high probability of causality and expressing causal semantics can be labeled as training data for the ECI task.

LearnDA (Zuo et al., 2021b) treats sentence-to-relation mapping (the target task, ECI) and relation-to-sentence mapping (the augmentation task, sentence generation) as dual tasks and models the mutual relationship between them through dual learning.

BT-ESupCL (Li et al., 2025) is an event-level supervised contrastive learning method with back-translation augmentation. It incorporates an efficient compound back-translation augmentation approach with three parallel strategies, designed to mitigate data scarcity while enhancing expression diversity and reducing computational costs.

4.4.2. Context generation

KMMG (Liu et al., 2020) is an event mention masking generalization mechanism that replaces event mentions with masks, forcing the model to rely on contextual information for predictions and aiming to exploit event-agnostic clues for reasoning.

SemDI (Li et al., 2024) is a heuristic semantic dependency inquiry network that first captures semantic dependencies using a unified encoder, then randomly masks one event from the event pair and utilizes a cloze analyzer to generate a fill-in token based on comprehensive context understanding. Finally, this fill-in token is used to inquire about the causal relationship between the two events in the given sentence.

GenSORL (Chen et al., 2024a) is a strategy gradient approach that optimizes the generation of dependency paths, facilitating the extraction of important contextual words for causality identification between event mentions. This, in turn, helps better capture the fundamental relevant knowledge between event mentions.

4.4.3. Representation learning

CauSeRL (Zuo et al., 2021a) is a self-supervised representation learning framework designed to learn context-specific causal patterns from external causal statements. This approach is motivated by the idea that the commonalities learned from different causal statements reflect context-specific causal patterns, which are helpful for identifying event causalities in unseen cases.

ECLEP (Pu et al., 2023a) is a framework that introduces event causal label information and event pair interaction information to enrich the representation of event pairs. This approach designs an event-causal-label-aware module to model the event causal label information and an event pair interaction graph module to model the interaction information between event pairs.

[2] https://platform.openai.com

[3] https://github.com/huggingface/peft

Table 2
Evaluation performance on the ESL, ESL♣ and CTB dataset. P, R and F1 indicate Precision, Recall and F1-score, respectively. The **best** results are highlighted in bold, and the second-best results are underlined. The symbol * denotes significance at the level of $p \leq 0.05$.

Categories	Models	ESL			ESL♣			CTB		
		P	R	F1	P	R	F1	P	R	F1
Data Augmentation	KnowDis (Zuo et al., 2020)	39.7	66.5	49.7	–	–	–	42.3	60.5	49.8
	LearnDA (Zuo et al., 2021b)	42.2	69.8	52.6	–	–	–	41.9	68.0	51.9
	BT-ESupCL (Li et al., 2025)	46.5	64.4	53.9	–	–	–	53.4	57.7	55.4
Context Generation	KMMG (Liu et al., 2020)	41.9	62.5	50.1	–	–	–	36.6	55.6	44.1
	SemDI (Li et al., 2024)	56.7	68.6	62.0	75.0	75.7	75.3	59.3	**77.8**	67.0
	GenSORL (Chen et al., 2024a)	65.6	63.3	64.6	–	–	–	66.2	57.0	60.9
Representation Learning	CauSeRL (Zuo et al., 2021a)	41.9	69.0	52.1	–	–	–	43.6	68.1	53.2
	ECLEP (Pu et al., 2023a)	49.3	68.1	57.1	–	–	–	50.6	63.4	56.3
	C3Net (Gao et al., 2024)	55.1	68.6	61.1	–	–	–	54.5	67.7	60.4
Structure Knowledge	SemSIn (Hu et al., 2023)	50.5	63.0	56.1	64.2	65.7	64.9	52.3	65.8	58.3
	GCKAN (Ding et al., 2024)	50.9	60.6	55.3	–	–	–	52.2	60.7	56.1
	DFP (Huang et al., 2024)	55.9	**69.8**	62.1	–	–	–	53.7	64.2	58.5
Prompt Tuning	KEPT (Liu et al., 2023)	50.0	68.8	57.9	–	–	–	48.2	60.0	53.5
	KIGP (Hu et al., 2025)	62.4	68.5	65.3	–	–	–	61.3	63.4	62.3
In-Context Learning	CarICL (Ours)	**66.8**	69.1	**67.9***	**77.3**	**77.3**	**77.3***	**68.9**	70.1	**69.5***

C3Net (Gao et al., 2024) is a competitive selection mechanism that utilizes representations of private evidence and historical patterns, generated by intuitive and logical causal reasoners, to determine which reasoner's prediction is more accurate for a given pair of events.

4.4.4. Structure knowledge

SemSIn (Hu et al., 2023) is a semantic structure integration model that employs abstract meaning representation parsers to convert unstructured text into an event-centric structure graph and an event-associated structure graph, and then integrates these two types of information using an attention mechanism.

GCKAN (Ding et al., 2024) is a graph contrast-based knowledge augmented network that designs a descriptive knowledge augmentation module to aggregate one-hop neighbor information from knowledge graphs for learning meaningful descriptive knowledge of events and a relational knowledge augmentation module to encode multi-hop path information and learn latent reasoning knowledge between event pairs.

DFP (Huang et al., 2024) is a pipeline that first distills a commonsense graph into a task-specific meta-graph and then fuses the interactions and relations between event mentions within the meta-graph. Finally, continual pre-training is proposed to further align and fuse the event mentions with the commonsense meta-graph.

4.4.5. Prompt tuning

KEPT (Liu et al., 2023) is a knowledge-enhanced prompt tuning framework that leverages prompt tuning to incorporate background information and relational information from external knowledge bases for causal reasoning.

KIGP (Hu et al., 2025) is a prompt tuning method guided by a knowledge interaction graph, designed to maximize the potential of PLMs for ECI by effectively integrating external knowledge through knowledge interaction.

4.4.6. Large language model

Text-Davinci-002, **Text-Davinci-003**, **GPT-3.5-Turbo**, and **GPT-4.0** are four progressive state-of-the-art versions of LLMs. These experiments are conducted under a zero-shot setting because previous studies (Gao et al., 2023; Li et al., 2024) found that the ICL paradigm and CoT prompts can further exacerbate the causal hallucination of LLMs, resulting in low accuracy.

To conduct a comprehensive evaluation, we also introduce open-source models such as Llama-2-7B-chat (Touvron et al., 2023). Following a previous study (Li et al., 2024), we define **Llama-2-ZeroShot** and **Llama-2-FineTune** as two versions of Llama-2-7B-chat, corresponding to the zero-shot and supervised fine-tuning settings, respectively.

4.5. Main result

Table 2 presents the performance of state-of-the-art models on the ESL, ESL♣, and CTB datasets. A key observation is that our proposed model, CarICL, outperforms existing PLM-based models for sentence-level ECI across these datasets. Specifically, it achieves improvements in F1-score of 2.6 %, 2.0 %, and 2.5 % over the state-of-the-art models, respectively. These improvements are statistically significant ($p \leq 0.05$), demonstrating the stability and consistency of CarICL and highlighting its advantages for the ECI task. Importantly, CarICL attains state-of-the-art performance on widely used ECI datasets without relying on additional annotations (e.g.,

Table 3
Evaluation performance on the ESL, ESL♣ and CTB dataset. We use the Δ metric to measure causal hallucination, i.e., $\Delta = |P - R|$. We use the #Params metric to measure the number of parameters in the backbone network. The symbol ↓ indicates that a lower value for this metric is better. The symbol † means that the data is sourced from Abacha et al. (2024).

Models	#Params ↓	ESL				ESL♣				CTB			
		P	R	F1	Δ ↓	P	R	F1	Δ ↓	P	R	F1	Δ ↓
Text-Davinci-002 (Gao et al., 2023)	175B†	23.2	80.0	36.0	56.8	–	–	–	–	5.0	75.2	9.3	70.2
Text-Davinci-003 (Gao et al., 2023)	175B†	33.2	74.4	45.9	41.2	–	–	–	–	8.5	64.4	15.0	55.9
GPT-3.5-Turbo (Gao et al., 2023)	175B†	27.6	80.2	41.0	52.6	–	–	–	–	6.9	82.6	12.8	75.7
GPT-4.0 (Li et al., 2024)	200B†	30.7	**85.7**	45.2	55.0	31.2	**86.3**	45.8	55.1	4.6	**84.6**	8.7	80.0
Llama-2-ZeroShot (Li et al., 2024)	7B	11.4	50.0	18.6	38.6	12.1	50.7	19.5	38.6	5.4	53.9	9.8	48.5
Llama-2-FineTune (Li et al., 2024)	7B	20.5	57.1	29.8	36.6	20.3	57.6	30.0	37.3	10.5	61.8	17.9	51.3
CarICL (Ours)	**7B**	**66.8**	69.1	**67.9***	**2.3**	**77.3**	77.3	**77.3***	**0.0**	**68.9**	70.1	**69.5***	**1.2**

Table 4
Ablation study on our proposed CarICL. The symbol ✓ indicates that the component is selected. To show that CarICL is a flexible framework (also effective in prompt-only settings), we replace the backbone model with GPT-3.5-Turbo (**ID 7**), which relies solely on our proposed prompt construction method (as shown in Fig. 4).

ID	CarICL			Backbone	ESL				ESL♣				CTB			
	Retrieval	Reasoning	Optimization		P	R	F1	Δ ↓	P	R	F1	Δ ↓	P	R	F1	Δ ↓
1	✓	✓	✓	Qwen-2.5-7B	**66.8**	69.1	**67.9***	**2.3**	**77.3**	77.3	**77.3***	**0.0**	**68.9**	70.1	**69.5***	**1.2**
2		✓	✓	Qwen-2.5-7B	56.3	62.4	59.2	6.1	67.1	72.3	69.6	5.2	55.2	64.3	59.4	9.1
3	✓		✓	Qwen-2.5-7B	60.2	64.0	62.0	3.8	69.6	72.5	71.0	2.9	61.0	67.3	64.0	6.3
4	✓	✓		Qwen-2.5-7B	50.2	63.3	56.0	13.0	63.1	74.2	68.2	11.0	53.3	65.8	58.9	12.4
5				Qwen-2.5-7B	21.9	43.5	29.1	21.6	19.6	59.6	29.5	40.0	4.1	36.9	7.4	32.8
6			Fine-Tuned	Qwen-2.5-7B	33.7	53.2	41.3	19.5	29.3	68.2	41.0	38.9	10.0	46.2	16.5	36.2
7	✓	✓		GPT-3.5-Turbo	58.2	**77.4**	66.4	19.2	68.0	**87.7**	76.6	19.8	56.7	**84.9**	68.0	28.2

external knowledge bases) or third-party tools (e.g., semantic dependency parsers). This contrasts with recent advancements in ECI, such as KIGP (Hu et al., 2025), which depends on human-annotated external knowledge bases, and SemDI (Li et al., 2024), which utilizes semantic dependency parsing tools. This distinction underscores that while LLM-based methods are effective, they have been underutilized in prior work. Our proposed CarICL effectively leverages the capabilities of LLM backbones, achieving superior results without the need for supplementary resources.

Moreover, Table 3 provides a comparison between our proposed CarICL and state-of-the-art LLM-based methods across all benchmarks, demonstrating that CarICL mitigates the causal hallucinations of LLMs. The most important observation from the table is that CarICL achieves significantly better performance than state-of-the-art LLM-based methods across three critical metrics: F1-score (F1), backbone parameters (#Params), and causal hallucination (Δ). First, it achieves F1-score improvements of 22.0 %, 31.5 %, and 51.6 % over the second-best models, respectively. These performance enhancements are statistically significant with $p \leq 0.05$, underscoring the clear advantages of CarICL in effectively addressing causal hallucination. Second, CarICL achieves better performance than state-of-the-art LLM-based methods with 175B or even 200B parameters while using 7B parameters. The high efficiency of CarICL makes it highly suitable for deployment in resource-limited and time-sensitive scenarios compared to the aforementioned LLM-based methods. Last, it reduces causal hallucination by 34.3 %, 37.3 %, and 39.1 % compared to the second-best models, respectively. We observe that baseline LLMs-based methods frequently display overconfidence in identifying causal relationships, leading to high recall but low precision. This observation aligns with previous research findings (Mielke et al., 2022; Xiong et al., 2024), referred to as the causal hallucination phenomenon in the ECI task (Chen et al., 2024b; Gao et al., 2023; Huang et al., 2024; Li et al., 2024).

5. Analysis

In this section, we further analyze the proposed methods: causality-aware demonstration retrieval, cause-and-effect reasoning, and causality preference optimization.

5.1. Ablation study

To examine the contributions of each component in our proposed CarICL framework, we conduct a comprehensive ablation study across three ECI datasets, with results summarized in Table 4. Each row corresponds to a variant of CarICL, where the presence or absence of three core components is explicitly controlled. Specifically, the three components — Causality-aware Demonstration Retrieval, Cause-and-Effect Reasoning, and Causality Preference Optimization — are denoted as “Retrieval”, “Reasoning”, and “Optimization”, respectively.

- Full model **(ID 1)** achieves the best performance across all datasets, validating the effectiveness of integrating all three components.

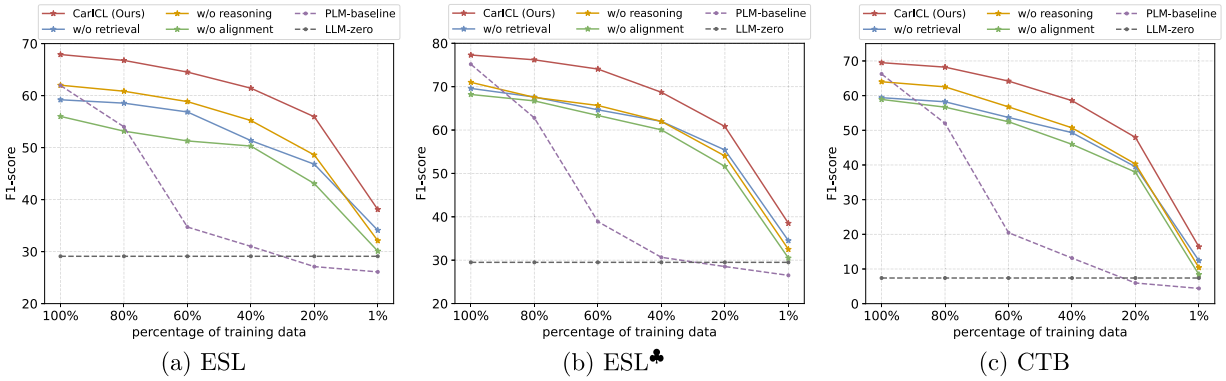


Fig. 5. Visualization of performance on the ESL, ESL♣, and CTB datasets in low-resource scenarios.

- Replacing Causality-aware Demonstration Retrieval with random demonstration retrieval **(ID 2)** has a notable impact on F1-score (dropping by 8.7 %, 7.7 %, and 10.1 %) and Δ (worsening by 3.8 %, 5.2 %, and 7.9 %). This may be because the LLM becomes more prone to confidently misidentifying causal pairs, leading to overconfidence. Removing the retrieval step further exacerbates the imbalance between causal and non-causal predictions.
- Removing Cause-and-Effect Reasoning **(ID 3)** results in F1-score drops of 5.9 %, 6.3 %, and 5.5 % across the three datasets. This suggests that vanilla ICL may only elicit shallow pattern-based reasoning, whereas our explicit step-by-step reasoning guide the LLM to uncover deeper causal structures.
- Excluding Causality Preference Optimization **(ID 4)** causes F1-score drops of 11.9 %, 9.1 %, and 10.6 %, and worsens Δ by 10.7 %, 11.0 %, and 11.2 %, respectively. This can be attributed to the limited causal knowledge in the small LLM, while our optimization component effectively injects comprehensive causal knowledge into the model.
- In contrast, **ID 5** and **ID 6** correspond to the zero-shot and fine-tuned versions of Qwen-2.5-7B, respectively — representing the backbone model without any of the proposed components. These configurations result in substantial performance degradation and exhibit severe causal hallucinations, confirming that **CarICL's effectiveness does not stem from domain-specific fine-tuning alone, but rather from its principled causal design**.
- We introduce a CoT-style prompting variant (**ID 7**) that relies solely on our proposed prompt construction method (as shown in Fig. 4) to show that CarICL is a flexible framework effective in both Causality Preference Optimization (ID 1) and prompt-only settings. It achieves F1-scores of 66.4 %, 76.6 %, and 68.0 %, outperforming the second-best ECI baselines — KIGP (65.3 %) and SemDI (75.3 %, 67.0 %). These results demonstrate that even without fine-tuning, our prompt-only method still delivers state-of-the-art performance.

5.2. Low-resource scenario

To demonstrate that our proposed CarICL has superior few-shot learning capabilities, we evaluate its performance in low-resource scenarios. The experiments are conducted by randomly sampling 80 %, 60 %, 40 %, 20 %, and 1 % of the training data for both fine-tuning and retrieval in both competitors. To ensure the replicability of our results, we fix the random seed at 42. We also remove specific components of CarICL and evaluate the performance of the resulting incomplete model on the ESL, ESL♣, and CTB datasets to assess the impact of our proposed methods. Following Section 5.1, we first replace the proposed causality-aware demonstration retrieval method with random demonstration retrieval, denoted as **w/o retrieval**. Next, we remove the proposed cause-and-effect reasoning method, denoted as **w/o reasoning**. Finally, we eliminate the proposed causality preference optimization method, denoted as **w/o alignment**. Additionally, we replicate the performance of SemDI (Li et al., 2024), the state-of-the-art PLM-based model (denoted as **PLM-baseline**), and reproduce Qwen-2.5-7B-Instruct (Yang et al., 2024a), the zero-shot LLM-based baseline (denoted as **LLM-zero**), to serve as two benchmarks for comparison.

Taking a close observation of Fig. 5 from left to right, we can find some insightful phenomena. First, the performance of the PLM-baseline drops significantly when the percentage of training data is reduced to around 60 %, whereas the performance of CarICL and its variants (w/o retrieval, w/o reasoning, and w/o alignment) remains relatively stable. This phenomenon can be attributed to the strong few-shot learning capabilities of LLM-based methods, confirming the findings of Chan et al. (2024), Chung et al. (2024), Qin et al. (2023). In contrast, PLM-based approaches are more data-intensive and often constrained by data scarcity issues (Cao et al., 2021; Li et al., 2025; Zuo et al., 2021a,b, 2020). Second, the performance of the LLM-zero approach is not affected by the reduction in training data, whereas the performance of the PLM-baseline method drops below that of LLM-zero when the training data is reduced to around 30 %. This phenomenon can be attributed to the abundant prior knowledge embedded in LLMs, which provides LLM-based methods with a more robust foundation for understanding causal relationships. Third, although the performance of CarICL gradually decreases as the percentage of training data decreases, the decline is relatively slower than that of other baselines. This phenomenon highlights the effectiveness of our proposed methods in fully harnessing causal hallucination to mitigate the causal data scarcity problem, ultimately enhancing LLMs' performance on the ECI task. Moreover, this phenomenon also indicates that the success of

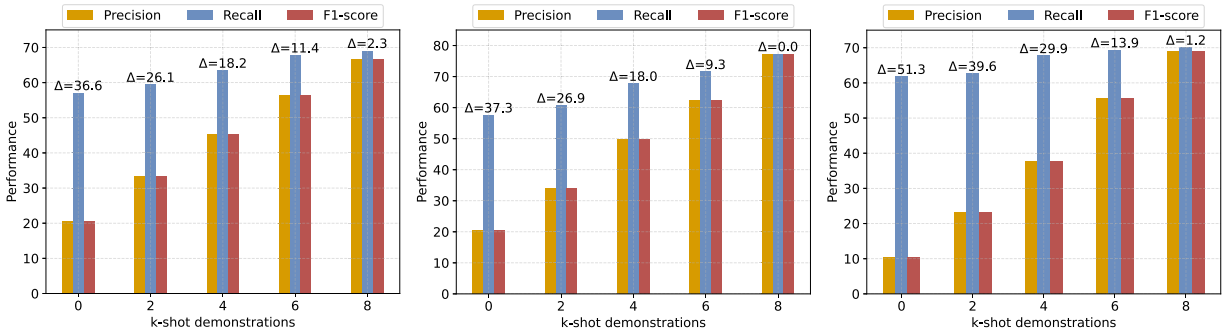


Fig. 6. Visualization of mitigating causal hallucination by increasing the number of demonstrations in our proposed CarICL. The experiment is conducted on the ESL, ESL♣, and CTB datasets.

Table 5
Evaluation performance of cross-task adaptation is measured by training on the ESL datasets and testing on the SET8 dataset. The symbol * denotes significance at the 0.05 level. Following Gao et al. (2023), Zou et al. (2025), we use the Δ metric ($\Delta = |P - R|$) to reflect causal hallucination in LLM-based methods. The symbol ↓ indicates that a lower value for this metric is better.

Methods	ESL → SET8			
	P	R	F1	Δ ↓
BERT (Devlin et al., 2019)	51.5	71.9	60.0*	−/−
KMMG (Liu et al., 2020)	59.4	75.0	66.0	−/−
KEPT (Liu et al., 2023)	57.8	80.4	67.2*	−/−
GCKAN (Ding et al., 2024)	58.3	80.6	67.7*	−/−
C3Net (Gao et al., 2024)	64.3	88.1	74.3	−/−
GPT-3.5-Turbo	13.2	**99.4**	23.3	86.2
Qwen-2.5-ICL	27.4	77.7	40.5	50.3
Qwen-2.5-FineTune	49.6	78.3	60.7	28.7
CarICL (Ours)	**67.0**	89.4	**76.6***	**22.4**

CarICL does not overly rely on finding high-quality causal demonstrations, making it highly effective in low-resource settings and beyond previous representative ICL works (Liu et al., 2022; Wan et al., 2023; Wang et al., 2024).

5.3. Visualization of mitigating causal hallucination

To demonstrate that our proposed CarICL can effectively mitigate the causal hallucinations of LLMs, we present its performance across several k-shot settings on the ESL, ESL♣, and CTB datasets. Specifically, we first determine the number of demonstrations, k, sampled from $\{0, 2, 4, 6, 8\}$. Then, we ensure that both the proposed causality preference optimization (Section 3.4) and the ICL inference stages use the same number of demonstrations for training and inference. The horizontal axis of Fig. 6 represents the number of demonstrations, while the vertical axis shows the performance of CarICL in terms of Precision, Recall, and F1-score. Moreover, we introduce Δ metric (as described in Section 4.2) to reflect causal hallucination and further demonstrate the effectiveness of CarICL. We can observe that as the number of retrieved causally similar demonstrations increases, the value of P increases, while the metric Δ decreases. This contrasts with vanilla ICL methods (Chen et al., 2024b; Gao et al., 2023; Huang et al., 2024; Li et al., 2024), where increasing the number of ICL demonstrations can further exacerbate causal hallucinations. These findings indicate that causal hallucinations can be progressively mitigated as the number of causally similar demonstrations increases, and that CarICL effectively reduces causal hallucinations by retrieving high-quality demonstrations.

5.4. Cross-task adaptation

To remain consistent with prior studies (Ding et al., 2024; Gao et al., 2024; Liu et al., 2020, 2023), we evaluate the generalization ability of our proposed CarICL by conducting a standard cross-task adaptation experiment. Specifically, we train the model on the **event** causality dataset ESL and test it on the SET8 dataset, which focuses on identifying causal relations between **entity** pairs. This setup, denoted as "ESL → SET8", has been widely used in previous work to assess whether a model can transfer its learned causal reasoning ability beyond event-centric domains. Compared to ESL, SET8 presents several notable challenges: it involves entity

Table 6
Evaluation of cross-domain adaptation.

Methods	HP			
	P	R	F1	Δ ↓
BiLSTM (Akkasi & Moens, 2021)	47.0	30.0	37.0	−/−
BioBERT (Akkasi & Moens, 2021)	52.0	51.0	51.0	−/−
MKFN (Hao et al., 2023)	62.4	48.9	54.8	−/−
GECANet (Zhang et al., 2023a)	64.4	61.1	62.7	−/−
MoRAG-FD (Zhao et al., 2024)	**67.1**	62.2	64.6	−/−
GPT-3.5-Turbo	23.0	**99.9**	37.4	77.0
Qwen-2.5-ZeroShot	26.3	28.4	27.3	2.1
Qwen-2.5-FineTune	37.3	39.2	38.2	1.9
CarICL (Ours)	66.8	66.8	**66.8***	**0.0**

nominals rather than event mentions, includes nine semantic relation types (only one of which is causality), and is derived from web-sourced, pattern-based sentence extraction. These differences provide a useful benchmark for evaluating the model's ability to generalize to out-of-distribution data.

For a thorough comparison, we evaluate the zero-shot performance of a strong LLM (e.g., GPT-3.5-Turbo) on the SET8 dataset. Following Gao et al. (2023), Zou et al. (2025), we use the Δ metric to specifically reflect causal hallucinations in LLM-based methods, while marking PLM-based methods as "−/−". For ablation purpose, we include **Qwen-2.5-ICL** and **Qwen-2.5-FineTune**, which represent two baseline configurations of Qwen-2.5-7B-Instruct (the backbone of CarICL) under standard in-context learning and supervised fine-tuning settings, respectively. Specifically, Qwen-2.5-ICL performs zero-shot inference on SET8 by retrieving demonstrations from ESL, while Qwen-2.5-FineTune is trained on ESL and then evaluated on SET8. Despite the serious class imbalance issue in the SET8 dataset, we do not conduct experiments under the dataset balancing setting (Liu et al., 2023).

Experiment results are listed in Table 5 and we can observe that: (1) While most models demonstrate high recall, they tend to have relatively low precision. In response, Liu et al. (2023) attribute the difficulties of PLM-based fine-tuning methods to class imbalance, whereas Gao et al. (2023), Zou et al. (2025) link the challenges faced by LLM-based zero-shot methods to causal hallucination. (2) Both Qwen-2.5-ICL and Qwen-2.5-FineTune show limited effectiveness in cross-task scenarios, highlighting the insufficiency of relying exclusively on vanilla LLM-based methods. (3) Our proposed CarICL outperforms all baseline models, demonstrating its ability to generalize causal reasoning beyond the event causality domain and across different semantic units (from events to entities).

5.5. Cross-domain adaptation

To further demonstrate the superior generalization ability of our proposed CarICL, we evaluate its performance on the biomedical ECI task. We first trained and evaluated the model on the **biomedical** ECI dataset, and then compared its performance with **biomedical-specific** ECI models. Despite the severe class imbalance in the HP dataset, we do not apply over-sampling techniques, unlike previous work (Liang et al., 2022). To validate the effectiveness of CarICL, we include state-of-the-art, biomedical-specific ECI models, such as MoRAG-FD (Zhao et al., 2024), which utilizes Retrieval-Augmented Generation (RAG) for training data augmentation through biomedical knowledge selection. To ensure a comprehensive comparison, we assess the zero-shot performance of a powerful LLM (e.g., GPT-3.5-Turbo) on the HP dataset. For ablation purpose, we also include **Qwen-2.5-ZeroShot** and **Qwen-2.5-FineTune**, which represent two baseline configurations of Qwen-2.5-7B-Instruct (the backbone of CarICL) under zero-shot inference and supervised fine-tuning settings, respectively.

The experimental results are presented in Table 6. **Firstly**, among biomedical-specific ECI baselines, MoRAG-FD remains the previous state-of-the-art, achieving a high precision (67.1 %) and recall (62.2 %), leading to a strong F1 score of 64.6 %. **Secondly**, GPT-3.5-Turbo encounters notable difficulties due to causal hallucination, achieving near-perfect recall (99.9 %) but exhibiting low precision (23.0 %). This leads to a suboptimal F1 score of 37.4 % and a high Δ value of 77.0 %, highlighting the severity of the causal hallucination issue. In comparison, Qwen-2.5-ZeroShot and Qwen-2.5-FineTune serve as the backbone models for CarICL; their relatively low scores indicate that our performance gains are not simply the result of domain-specific fine-tuning. **Finally**, CarICL achieves a significant milestone with an outstanding balance of precision, recall, and F1 score (66.8 %), coupled with a Δ value of 0.0 %, indicating minimal influence from causal hallucination. This demonstrates that CarICL not only adapts well to biomedical ECI but also exhibits impressive cross-domain adaptability, making it a powerful ECI method across various fields.

5.6. Alternative approaches to cause-and-effect reasoning

To explore alternative approaches for generating the proposed Cause-and-Effect Reasoning, we evaluate the overall performance of the proposed CarICL under different reasoner settings on the ESL, ESL♣, and CTB datasets. Since open-source LLMs can be easily deployed locally, thereby improving the reproducibility of CarICL, we introduce two widely used models (Qwen-2.5-72B-Instruct and Qwen-2.5-7B-Instruct) to replace the previously used closed-source third-party API-based reasoner, GPT-3.5-Turbo. The experimental results are presented in Table 7, and we observe the following phenomena: First, although Qwen-2.5-72B-Instruct has only 72 billion parameters, significantly fewer than the 175 billion Abacha et al. (2024) parameters of GPT-3.5-Turbo, it still achieves comparable

Table 7
Performance evaluation of alternative approaches to Cause-and-Effect Reasoning. The symbol * denotes statistical significance at the 0.05 level.

Models	ESL				ESL♣				CTB			
	P	R	F1	Δ ↓	P	R	F1	Δ ↓	P	R	F1	Δ ↓
CarICL (Ours)												
GPT-3.5-Turbo	66.8	69.1	67.9*	2.3	77.3	77.3	77.3*	0.0	68.9	70.1	69.5*	1.2
Qwen-2.5-72B-Instruct	65.8	69.7	67.7*	3.9	76.8	77.6	77.2*	0.8	68.1	70.4	69.2*	2.4
Qwen-2.5-7B-Instruct	64.5	68.6	66.5*	4.1	74.8	77.8	76.3*	3.0	66.5	70.0	68.2*	3.5

Table 8
Qualitative evaluation of cause-and-effect reasoning. We assess both logical validity and task relevance. The event mentions in the sentence are underlined.

***Sentence 1*:**
A firebomb attack on a bank in Greece killed at least three people today as police fought pitched battles with striking protesters furious at brutal budget cuts designed to avoid national bankruptcy.
***Rationale*:**
The word "attack" implies an intentional act of violence, which can cause harm or death. In this case, the firebomb attack on the bank resulted in the death of at least three people, establishing a causal relationship between the attack and the deaths.
***Ground Truth*:** Causality. ***Logical validity*:** High. ***Task Relevance*:** High.

***Sentence 2*:**
The Texas department of criminal justice has made sweeping leadership changes at a Huntsville prison where an inmate managed to get a gun that he later used to escape, the agency announced Tuesday.
***Rationale*:**
The causal relationship between the words "changes" and "escape" is that the leadership changes made by the Texas Department of Criminal Justice at the Huntsville prison led to an inmate being able to get a gun and escape.
***Ground Truth*:** Causality. ***Logical validity*:** Low. ***Task Relevance*:** High.

***Sentence 3*:**
Mr. Bush forestalled further diplomatic maneuvering by issuing an ultimatum on behalf of the allies, demanding that Iraq withdraw within a week, starting at noon on Saturday.
***Rationale*:**
The words "demanding" and "forestalled" do not have a causal relationship because issuing an ultimatum does not directly cause the prevention of further diplomatic maneuvering.
***Ground Truth*:** Non-Causality. ***Logical validity*:** High. ***Task Relevance*:** High.

performance. Second, as the number of LLM parameters decreases, the value of Δ gradually increases. CarICL still achieves satisfactory performance in terms of F1-score when leveraging the 7-billion-parameter LLM, Qwen-2.5-7B-Instruct. The above phenomena suggest that the proposed CarICL has excellent robustness, that is, it does not heavily rely on any particular LLM.

5.7. Qualitative analysis of cause-and-effect reasoning

To assess the quality of the generated cause-and-effect reasoning, we conduct a qualitative analysis examining the logical validity and task relevance of the generated rationales (as described in Section 3.3.2). We randomly selected three examples from the ESL and CTB datasets, covering a variety of event types and model outcomes. For each example, we annotated the logical validity (i.e., whether the causal explanation is coherent and grounded in real-world logic) and task relevance (i.e., whether the rationale focuses on the given causal relation rather than unrelated background knowledge). Based on the examples in Table 8, we make the following observations: The employed LLM is capable of producing logically sound and contextually relevant rationales when causal links are explicit (e.g., "attack" → "killed"). It may produce hallucinated causal reasoning even when the underlying connection is weak or speculative (e.g., inferring that "leadership changes" caused an "escape"). It can correctly identify non-causal cases and generate faithful negation rationales (e.g., "forestalled" vs. "demanding"), suggesting an ability to reason negatively as well. These analyses highlights avenues for improvement in CarICL's reasoning method. There is a need for mechanisms to verify causal coherence, especially for instances where causal inference requires implicit background knowledge. Enhancing the employed LLM with commonsense or structural constraints may help reduce reasoning hallucinations.

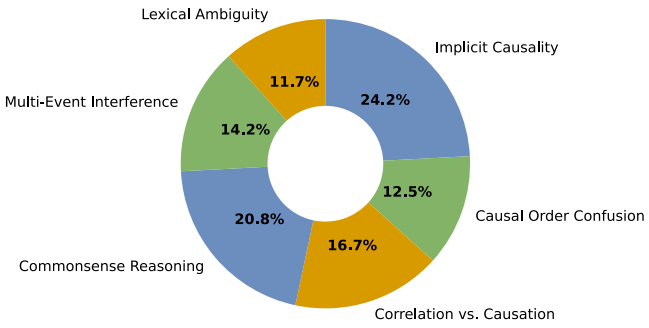


Fig. 7. Statistics of error types.

5.8. Error analysis

To better understand the limitations of our proposed CarICL and to identify concrete directions for future improvements, we conducted a fine-grained error analysis focusing on specific types of event relations where the model tends to struggle. We manually annotated 120 randomly selected error cases from CarICL's predictions across the ESL, ESL♣ and CTB datasets. As shown in Fig. 7, each case was categorized into one of six major error types based on the nature of the event relations and the underlying reason for the model's failure.

Implicit causality (24.2 %). These instances require the model to reason beyond surface-level cues and make inferences grounded in real-world knowledge. For example, in the sentence: "Greece's parliament is due to vote on the hefty cuts and reforms proposed by the government to address the country's financial crisis", the causal relation is "cuts → crisis". Although no explicit causal marker is present, it is commonsensical that the economic crisis is the motivating factor behind the fiscal reforms.

Causal order confusion (12.5 %). Some errors arise from difficulty in distinguishing cause from effect, especially when events occur in close proximity or are embedded in ambiguous syntactic constructions. For example, in the sentence "His supervision was revoked and he returned to prison in May 1984 with a new charge of indecency with a child", the causality is "charge → revoked". Although the charge and revocation co-occur, the direction of causality is either unclear or not explicitly expressed.

Correlation vs. causation (16.7 %). These types of errors are particularly common in descriptions where multiple events share a context but lack a direct causal link. For example, in the sentence "The outage initially affected all SEACOM traffic to Europe", the causality is "affected → outage". However, the outage is the cause, not the effect-both events are part of a single process.

Commonsense reasoning (20.8 %). Some cases reflect difficulties in applying basic commonsense reasoning, especially regarding psychological motivations or expected behavioral outcomes. For example, in the sentence "He had a history of mental health problems and had pleaded in vain to be kept locked up because he feared harming himself or someone else.", the causality is "locked up → harming". Despite the clear motivational intent expressed in the sentence, the model fails to capture the psychological cause-and-effect relationship between fear and the request to be detained.

Multi-event interference (14.2 %). In complex sentences with multiple events, the model needs to identify the correct causal chain, especially when events are indirectly linked across clauses or sentence boundaries. For example, in the sentence "Police intervened after more than 100 people gathered for a candlelit vigil in the East Flatbush neighborhood of Brooklyn, New York, where 16-year-old Kimani Gray was shot multiple times by police on Saturday.", the causality is "shot → gathered". The causal connection spans multiple clauses and is indirect: the shooting caused the vigil.

Lexical ambiguity (11.7 %). Ambiguous or figurative language may mislead the model, particularly when words are used in ironic or non-literal ways. For example, in the sentence "It was probably chance when the Wasilla Bible Church was burned, and arson was suspected.", the causality is "arson → chance". The phrase "probably chance" is likely used ironically, and "chance" does not literally cause "arson".

This analysis reveals that scenarios requiring deep reasoning, commonsense inference, or multi-hop event tracking are still challenging. These observations point to promising directions for future research, including enhancing temporal and discourse-level modeling, integrating external knowledge, and improving semantic robustness.

5.9. Computational overhead analysis

While our proposed CarICL introduces additional components beyond standard ICL, including causality-aware retrieval, cause-and-effect reasoning generation, and causality preference optimization, we design each method with attention to runtime efficiency

Table 9
Comparison of computational cost between our proposed CarICL and vanilla ICL. The symbols "+" and "≈" denote additional overhead relative to vanilla ICL and a negligible difference, respectively. The unit of measurement for computational overhead is seconds per sample. **All added costs are limited to the offline or training stages**.

Component	Stage	Cost type	CarICL	Standard ICL	Overhead
Retrieval Embedding	Offline	GPU compute (once)	8.53×10^{-5}	8.33×10^{-5}	≈
Retrieval Search	Inference	Vector similarity	5.67×10^{-6}	5.67×10^{-6}	≈
Reasoning Generation	Offline	ChatGPT API (once)	2.79	–	+
Preference Optimization	Training	Finetuning (once)	9.45	–	+
Prompt Assembly	Inference	Text concat	1.60×10^{-3}	1.60×10^{-3}	≈
Prompt Inference	Inference	1 forward pass	2.17×10^{-2}	2.17×10^{-2}	≈

and practical offline pre-processing trade-offs. To demonstrate this, we provide a detailed breakdown of the computational cost associated with each component of CarICL, in comparison to standard ICL, which is based on sentence-similarity retrieval. Table 9 summarizes the cost across different pipeline stages, measured in seconds per sample.

Offline and training cost. Compared to standard ICL, CarICL adds two non-inference-stage components. We generate cause-and-effect rationales for retrieved demonstrations using ChatGPT. This step incurs a one-time cost of 2.79 seconds per sample and is performed only during offline preprocessing. The generated rationales are reused across multiple prompts and evaluation runs, amortizing the cost. The proposed causality preference optimization introduces an additional training cost of 9.45 seconds per sample, incurred once during the ICL warm-up phase. It is not used at inference time.

Retrieval and inference cost. CarICL maintains nearly identical inference-time costs compared to standard ICL. Since both systems require embedding for candidates and queries, embedding computation takes 8.53×10^{-5} seconds per sample in CarICL — nearly the same as in standard ICL (8.33×10^{-5}) — and retrieval search incurs identical cost (5.67×10^{-6}). Prompt concatenation and forward pass latency remain unchanged (1.60×10^{-3} and 2.17×10^{-2} seconds per sample, respectively), thereby ensuring that CarICL maintains the same runtime characteristics as standard ICL during deployment.

All additional computational overhead introduced by CarICL occurs in the offline or training phase. Once the model is constructed, CarICL requires no additional reasoning generation or fine-tuning at test time. We think this cost profile offers a favorable trade-off given the substantial improvements in performance and reliability. Moreover, all added components are modular and can be optionally cached or shared across runs to reduce duplication.

6. Discussion

6.1. Theoretical implications

This study contributes to ECI and causal reasoning in NLP through four key insights. **First**, we propose CarICL, a novel ICL framework that mitigates causal hallucination by aligning LLM reasoning with human causal cognition, thereby deepening understanding of how chain-of-thought alignment and preference optimization jointly enhance contextual causal inference. **Second**, our proposed causality-aware demonstration retrieval extends vanilla ICL theory by prioritizing causal over semantic similarity, showing that ICL effectiveness depends on both semantic closeness and causal representativeness-an essential shift for reasoning-centered tasks. **Third**, integrating explicit cause-and-effect reasoning bridges surface pattern learning and interpretable causal understanding, guiding models to move beyond correlation toward structured reasoning. **Finally**, the proposed causality preference optimization advances theoretical understanding of preference learning by balancing causal and non-causal knowledge within LLMs. By mitigating reporting bias without disrupting existing representations, it provides a cognitive foundation for building causally reliable and interpretable language models.

6.2. Practical implications

The proposed CarICL framework also offers substantial practical implications for real-world applications. **First**, it demonstrates strong performance in low-resource scenarios, significantly reducing dependence on large-scale annotated datasets. This makes it particularly useful in domains such as healthcare, law, and social media analysis, where labeled causal data are often limited or costly to obtain. **Second**, by suppressing causal hallucinations, it improves both precision and robustness in automated causal inference, which is vital for high-stakes applications like clinical decision support, risk assessment, and legal reasoning. **Third**, its modular and domain-agnostic design enables broad deployment across text types (e.g., news, biomedical, scientific), ensuring strong generalization across in- and out-of-distribution data. **In summary**, CarICL enhances the causal reasoning capacity of LLMs through a scalable, interpretable, and efficient framework that effectively bridges theoretical innovation and practical deployment.

7. Conclusions and future works

In this paper, we propose a novel ICL framework that identifies event causalities while mitigating causal data scarcity and causal hallucination problems. Specifically, we propose (1) a causality-aware demonstrations retrieval method to search causally similar ICL examplars, (2) a cause-and-effect reasoning method to enhance the effectiveness of ICL inference, and (3) a causality preference optimization method to align LLMs with human causal cognition. Experiments demonstrate that our proposed CarICL not only outperforms state-of-the-art baselines but also surpass 175B LLMs with 7B parameters. Analyses reveal that: (1) by leveraging LLM and ICL techniques, CarICL can effectively handle low-resource scenarios and generalize to other tasks; and (2) CarICL can mitigate causal hallucinations by retrieving high-quality demonstrations. In the future, we will explore applying CarICL to zero-shot and few-shot setting for the ECI task.

CRediT authorship contribution statement

Yubo Feng: Conceptualization, Methodology, Software, Writing – original draft; **Lishuang Li:** Validation, Resources, Supervision, Project administration, Funding acquisition; **Xueyang Qin:** Validation, Writing – review & editing, Visualization; **Zuocheng Li:** Formal analysis, Investigation, Data curation.

Data availability

Data will be made available on request.

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